



Guidelines

Intravenous Iron

Reference #:MM009

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Medical, Nursing, Pharmacy,
Out of Scope	This guideline must NOT be used in the management of Haemodialysis patients: refer to the Medical Specialties Division Policy and Procedure: Anaemia management (Iron) in Haemodialysis Patients.	

Introduction

This guideline provides hospital specific guidance to all Sir Charles Gairdner Osborne Park Hospital Care Group (SCGOPHCG) staff involved in the prescribing, administration, storage or handling of intravenous iron within SCGOPHCG. This document supports and complies with related legislation, policy and other related documents [here](#).

1.0 General Information^{1,2}

Intravenous iron is available in a number of preparations with varying infusion rates. It is important to ensure the prescribed iron and iron being prepared and administered is the same.

IV Iron has restricted Inpatient use – refer to availability.

Intravenous (IV) iron products are indicated in the treatment of iron deficiency and anaemia when oral iron supplements cannot be given or have not worked. The prescribing, dosing, administration, and safety information differs between each IV iron product. For product specific details consult individual product information (available through MIMs).

It is important that iron stores be replaced before prescribing Erythropoietin Stimulating Agents (ESA's) such as Epoetin alfa, Darbepoetin alfa, or Epoetin beta. Contact Haematology consultant for appropriate timing of ESA following iron administration.

If the patient has active infection, consult with clinical microbiologist or infectious disease specialist before prescribing. (Preferable to have infection under control before initiating intravenous iron).

If the patient has received over 4 units of packed red blood cells (RBC) prior to iron infusion, ensure iron studies have been done post transfusion and **look at transferrin saturation only** (goal 20-30% transferrin saturation). Ferritin levels will **not be of value** during this post transfusion period.

An IV iron product must be used cautiously in patients with history of reaction to any other parenteral iron products. For more information on these processes please consult the

Patient Blood Management website on HealthPoint: <https://scgophcg-healthpoint.hdwa.health.wa.gov.au/directory/Surgical/PatientBloodManagement/Pages/default.aspx>

2.0 Indications^{1,3,4,5}

1. Absolute* or functional** iron deficiency.
2. Patients who are unable to tolerate oral iron.
3. Patients who are non-compliant with oral iron.
4. Oral iron is ineffective due to malabsorption, gastric surgery or other pre-existing medical conditions.

* Absolute iron deficiency is defined as ferritin <15-30mcg/L, or ferritin <100mcg/L with transferrin saturation <20%

** Functional iron deficiency exists when, despite adequate stores, iron cannot be mobilised for erythropoiesis. It is commonly seen in patients with end stage renal failure whose response to erythropoietin stimulating agents (ESAs) may be optimised when ferritin is >200mcg/L.

3.0 Contraindications^{1,3,4,5}

3.1 Absolute

1. Evidence of iron overload (transferrin saturation >45%).
2. Bacteraemia.
3. Decompensated hepatic disease or infectious hepatitis or infectious renal disease.
4. Pregnancy in the first trimester.

Pregnancy Note: see [Pregnancy](#) below which refers to the King Edward Memorial Hospital Guidelines.

3.2 Relative

1. Anaemia not attributed to iron deficiency.
2. Severe infection or inflammation including inflammatory arthritis.
3. History of significant allergy or asthma.
4. Polycythaemia vera.
5. Suspected reactions to IM or IV iron.
6. Uncontrolled hyperparathyroidism.
7. Ferric derisomaltose (Monofer[®]) is not recommended for use in children and adolescents < 18 years due to insufficient data on safety and efficacy.
8. Ferric carboxymaltose (Ferinject[®]) has not been studied in children and therefore is not recommended in children under 14 years.
9. Haemodialysis dependent chronic kidney disease - Refer to Anaemia management (Iron) in Haemodialysis Patients.

4.0 Pregnancy^{6,7,8}

Intravenous Iron is contraindicated in the 1st Trimester of pregnancy:

- Is your patient in second or third trimester of Pregnancy?
- Does your obstetric patient have iron deficiency anaemia?
 - Hb < 105g/L 2nd Trimester, Hb < 110g/L 3rd Trimester; **And**
 - Ferritin <30ug/L.

If yes to ALL of the above, and your patient is free from bacterial infection, consider prescribing Ferric Carboxymaltose (Ferinject[®]). NOTE: Restrictions exist for inpatient use of Ferric Carboxymaltose, please refer to them under [Availability](#).

For further advice and monitoring regarding iron in this population, please refer to both the KEMH monograph and clinical practice guideline below:

- <https://kemh.health.wa.gov.au/~media/Files/Hospitals/WNHS/For%20health%20professionals/Clinical%20guidelines/OG/WNHS.OG.IronTherapyIV.pdf>
- https://kemh.health.wa.gov.au/~media/Files/Hospitals/WNHS/For%20health%20professionals/Clinical%20guidelines/Pharmacy/medications/ferric_carboxymaltose.pdf

5.0 Safety Recommendations & Prescription of Iron^{2,3,9,10}

5.1 Prescription of Iron

- IV iron therapy is prescribed on the IV fluid chart (include dose, diluent, rate).
- In addition to the above, a PBS compliant prescription is required for outpatients and day-admitted patients (e.g. IV lounge, Cancer Centre and relevant site DPU).
- For OPH Orthopaedic Patients:
 - The SCGH Patient Blood Management (PBM) team determine the IV Iron dose the patient requires. The Anaesthetic Registrar has governance of the patient. Ferric Carboxymaltose is the formulation of choice for orthopaedics.
- For OPH Gynaecology and General Surgery Urology Patients:
 - The IV Iron dose and formulation is determined in consultation with OPH pharmacy. The Specialty Registrar has governance of the patient.

5.2 Oral iron therapy

- When possible, the patient should cease oral iron therapy one week prior to infusion and oral therapy should not be recommenced until one week after last infusion of iron.

5.3 Permanent Skin Staining and Patient consent¹⁰

Iron deposition can occur at sites of skin inflammation following infusion and cause widespread skin staining or hyperpigmentation at sites distant from the IV cannulation. This occurs due to increased vascular permeability in the skin due to inflammation. In patients with widespread skin rash, it is recommended that the infusion is deferred until rash or skin inflammation has subsided.

Prior to administering, patient consent must be obtained and documented. This conversation must include mention of adverse effects like permanent skin staining. A patient information sheet is available to assist with this (available on HealthPoint > Patient Blood Management [here](#)).

- **Skin staining as a result of extravasation is irreversible.** (Refer to section on [extravasation](#) at the end of this guideline).
 - Ensure the cannula is in the largest vein possible and secured.
 - Renal patients (dialysis and non-dialysis): no cannulation or blood sampling above the wrist.
 - Flush with 50mL sodium chloride **before and after iron** to minimise the risk of skin staining.

5.4 Recommendations for safe administration are:

- **Wherever possible**, iron infusions in **clinically stable** patients should be avoided between 2100-- 0700hrs at SCGH and between 1700 – 0700hrs at OPH, unless urgently required and medical staff are available.

- Once diluted, use as soon as practicable. If storage is necessary, store at 2-8C for up to 12 hours.
- An infusion pump must be used to administer iron infusions.
- Do not add any other medications to infusion or mix in the same line.
- Follow the hospital guidelines for checking the preparation and patient before commencing the infusion.
- Resuscitation equipment, including oxygen, adrenaline, hydrocortisone and promethazine must be readily available in the clinical area.
- Advise the patient to report adverse reactions or extravasation promptly.
- In case of a hypersensitivity reaction, treatment should be stopped immediately and appropriate management initiated.
- ACE inhibitors may increase the incidence of adverse effects like erythema, hypotension, nausea and vomiting.
- Prophylaxis against allergic reactions is unnecessary, except where there is a history of severe asthma/allergy or previous adverse reaction to parenteral iron. Consider a premedication of 10mg oral Promethazine 30 minutes prior to commencing infusion for these patients.

6.0 Availability^{3,4,5}

Intravenous Iron	Availability	Formulary Restrictions – As per the WA SME
1. Iron POLYmaltose complex (Ferrosig [®])	1gram/20mL 500mg/10mL	Unrestricted
2. Ferric CARBOXYmaltose (Ferinject [®])	500mg/10mL 100mg/2mL	1. Inpatient: For use in patients under the care of G75, MAU, ED or SSU who are also for urgent discharge (same day). 2. Outpatient or day admitted patients (e.g. IV lounge, Cancer Centre and relevant site DPU): For use when oral preparations are ineffective/not appropriate. 3. For patients with a documented hypersensitivity to iron POLYmaltose. To be supplied via Pharmacy on a named patient basis only.
3. Ferric DERISOmaltose (Monofer [®])	500mg/5mL	1. Inpatient: * This medication is restricted to outpatient and day-patients only. All inpatient initiation requires an Individual Patient Approval 2. Outpatient or day-admitted patients (e.g. IV lounge, Cancer Centre and relevant site DPU): • For patients requiring a dose > 1000 mg of elemental iron. • For patients who cannot tolerate ferric carboxymaltose (previous hypophosphatemia or mild to moderate hypersensitivity). To be supplied via Pharmacy on a named patient basis only.

7.0 OPH DPU Procedure

- For Orthopaedic Patients:
 - The SCGH Patient Blood Management (PBM) nurse and OPH Area Manager Surgical Wards and Clinics liaise and jointly determine day/time of admission.
 - The SCGH PBM Nurse advises the patient in writing of admission details.
 - The patient, or someone accompanying them, is to collect and pay for the medication from the pharmacy, located in J Block OPH at the time of the procedure.
- For all other patients:
 - The Speciality Registrar liaises and jointly determines day/time of admission and confirms who will inform the patient.
 - The Area Manager Surgical Wards and Clinics advises OPH relevant departments.

8.0 Dose, Preparation, Administration, Monitoring and Adverse Effects

The intravenous iron product should be given in accordance with the method of administration stated in the product information. IV iron should only be given in an environment where the patient can be adequately monitored, and where resuscitation facilities are available. See Dosage tables below.

Table 1. Iron POLYmaltose (Ferrosig®)^{1,2,3,6,7,11,12,13,14,15}

Iron POLYmaltose (Ferrosig®)
Formulary Restrictions: see restrictions under availability here
Dose
• Single dose should not exceed 2500mg • Recommended dose is based on body weight*, however for patients with iron deficiency anaemia, higher doses of iron may be required. ≤70kg is 1000mg >70kg is 1500mg • For patients with severe iron deficiency anaemia higher doses of iron may be required. • To calculate a total body iron infusion dose, please refer to the product information (MIMS) or contact your Clinical Pharmacist for advice. • *For overweight patients, an estimated healthy body weight should be used when considering doses.
Preparation
Prepared by Nursing Staff. • A filter needle should be used. • Protect from light. • Prime the line with the iron infusion to ensure delivery of the test dose in the first 15 minutes. • For doses up to 1500mg, the total dose of iron is added to 250mL Sodium Chloride 0.9%. • For doses greater than 1500mg the dose of iron is added to 500ml Sodium Chloride 0.9%.

Iron POLYmaltose (Ferrosig®)

Administration

Patients can be considered for an accelerated infusion rate* if they are haemodynamically stable and otherwise well.

Exclusions for accelerated infusion include:

- previous adverse reaction to iron infusion; or
- cardiac failure; or
- inflammatory disease states (inflammatory bowel disease); or
- dose > 1500mg.

Patient Criteria	Dilution and Administration
*Accelerated infusion rate for doses up to 1500mg AND patient haemodynamically stable and otherwise well • (see exclusion criteria)	• Dilute in 250mL Sodium Chloride 0.9%. • First 15 min: Commence infusion at 40mL/hr. • If there are no reactions after 15 minutes the rate can be increased to 190mL/hr. • Infusion duration approx. 90min.
Doses up to 1500mg	• Dilute in 250mL Sodium Chloride 0.9%. • First 15 min: Commence infusion at 40mL/hr. • If there are no reactions after 15 minutes the rate can be increased to 125mL/hr. • Infusion duration approx. 2¼ hrs.
Doses >1500mg-2500mg Single dose MUST NOT exceed 2500mg	• Dilute in 500mL Sodium Chloride 0.9%. • First 15 mins: Commence infusion at 40mL/hr. • If there are no reactions after 15 minutes the rate can be increased to 160mL/hr. • Infusion duration approx. 3½ hrs.

Reassess iron stores

- 14-21 days after last infusion.

Monitoring requirements

- **Prior to commencing infusion: Record baseline core physiological observations** Temp (T), Pulse (P), Resp Rate (R), Blood Pressure (BP) and Oxygen Saturation (SpO₂)
- **15 minutes after commencement of infusion:** Repeat observations. If there are no adverse events, increase the rate.
- Continue hourly observations for the duration of the infusion.
- **Patients should remain in the clinical area for the duration of the infusion plus 30 minutes post completion.**
- **30 min after completion of infusion:** Repeat observations

An identified medical officer should be readily available but there is no need to remain with the patient.

Iron POLYmaltose (Ferrosig®)**Possible adverse effects**

- Rash, urticaria, phlebitis
- Flushing, sweating, chills, fever
- Headache, dizziness, nausea, vomiting
- Syncope, tachycardia, hypotension, circulatory collapse
- Hypophosphataemia
- Bronchospasm, angioedema
- Late reaction of flu like illness with headache, aches, myalgia and fever can occur 2-5 days after infusion.
- **Anaphylaxis and hypersensitivity reactions are uncommon

Table 2. Ferric CARBOXYmaltose & Ferric DERISOmaltose^{1,2,4,5,7,15}

Ferric CARBOXYmaltose (Ferinject®)	Ferric DERISOmaltose (Monofer®)
Formulary Restrictions	
• Availability - See restrictions here. • Ferric Carboxymaltose is the preferred first line agent in the Outpatient Setting.	• Availability - See restrictions here. • Ferric Derismomaltose is the second line treatment choice in the outpatient setting.
Refer to the WA Statewide Medicines (SMF) for availability and prescribing restrictions for update to date prescribing restrictions of all formulations: https://formulary.hdwa.health.wa.gov.au/Home	
Dose	
• Single doses should not exceed 1000mg or 20mg/kg. • Recommended dose based on body *weight. ≤70kg is 500mg >70kg is 1000mg • For patients with severe iron deficiency anaemia, higher doses of iron may be required. • To calculate a total body iron infusion dose, please refer to the product information or contact your Clinical Pharmacist for advice. • The maximum single dose is 1000mg (or 20mg/kg if patient's weight is < 70kg) in one week. • For patients with an Hb value ≥ 140 g/L, an initial dose of 500 mg iron should be given, and iron parameters should be checked prior to repeat dosing.	• Single doses should not exceed 1500mg or 20mg/kg. • Recommended dose based on body *weight. ≤70kg is 1000mg >70kg is 1500mg • Ferric Derisomaltose doses should be capped at 1500mg. • For doses greater than 1500mg, repeat the infusion in seven days to give the remainder of total dose.

Ferric CARBOXYmaltose (Ferinject®)	Ferric DERISOmaltose (Monofer®)
- For extremes of body weight, consider using the Ganzoni method. If Ganzoni method is used and body weight: ≤ 66 kg round dose down to nearest 100mg 66 kg round dose up to nearest 100 mg - For calculated doses >1000mg give remainder of dose 7 days after initial infusion. Two 500mg doses can be administered in one week, no closer than 48 hours apart. *For overweight patients, an estimated healthy body weight should be used when considering doses	
Preparation	
Prepared by Nursing Staff <70kg 500mg in 100mL sodium chloride 0.9%. ≥70kg 1000 mg in 100mL sodium chloride 0.9%.	Prepared by Nursing Staff <70kg 1000mg in 250mL sodium chloride 0.9%. >70kg 1500mg in 250mL sodium chloride 0.9%.
Administration	
Infuse over 15 minutes	Infuse over 30 minutes
Reassess iron stores	
21-28 days after last infusion	21-28 days after last infusion
Monitoring requirements	
Prior to commencing infusion: Record baseline core physiological observations Temp (T), Pulse (P), Resp Rate (R), Blood Pressure (BP) and Oxygen Saturation (SpO₂). On completion of infusion: Repeat observations. Patients should remain in the clinical area for the duration of the infusion plus 30 minutes post completion. 30 min after completion of infusion: Repeat observations. An identified medical officer should be readily available but there is no need to remain with the patient.	
Possible adverse effects	
- Headache. - Dizziness. - Nausea, abdominal pain, constipation, diarrhoea. - Late reaction of flu like illness with headache, aches, myalgia and fever can occur 2-5 days after infusion. - Anaphylaxis and hypersensitivity reactions are rare.	

9.0 Extravasation^{10,15}

There is a risk that administration of all IV formulations of iron can cause extravasation; leakage of iron out of blood vessels into surrounding tissue. If redness or blistering is obvious, tissue can become necrotic and management is aimed at preventing further damage to tissue.

9.1 Signs and Symptoms of extravasation:

- Tenderness/discomfort at insertion site.
- Swelling above or below insertion site.
- Taut skin above or below insertion site.
- Fluid leak at insertion site.
- Coolness/blanching around insertion site.
- Numbness or tingling above or below insertion site Burning stinging pain.
- Redness may occur followed by blistering, tissue necrosis and ulceration.

If extravasation is suspected, follow the steps below.

9.2 Management of extravasation:

1. Stop infusion immediately and remove the cannula. The remainder of the IV iron infusion is abandoned.
2. If iron staining is immediately visible, measure the site and arrange for hospital photographs to be taken (organise with Audio Visual Production Unit (AVPU), for more information see the AVPU HealthPoint page [here](#)). This will aid ongoing monitoring of staining. Document the volume of infused fluid which may aid in assessing the volume of iron infiltrated.
3. Apply a cold pack to the infiltrated site and elevate the affected limb.
4. Reassure and provide a full explanation to the patient.
5. Inform the Medical Officer so an assessment can be made of sensory deficit which could indicate nerve damage or compartment syndrome.
6. Document the management in the medical records and complete a clinical incident form.
7. Follow-up as an outpatient where long-term management will be discussed if needed.
8. Further advice may be required from other specialities including Dermatology (skin staining), Plastic Surgery (sensory deficit) or Haematology (anaemia management). Laser therapy has been successful in reducing the skin staining long term.

Related Documents

Legislation:

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)

Authorising Policy and Standard/s:

- [WA Department of Health Consent to Treatment Policy 2016](#)
- [Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy](#)
- [SCGOPHCG Formulary Management Procedures MMP013](#)
- [SCGH Hospital Policy 141 Medicines Management](#)
- [SCGH Medicines Management Procedures and Standards](#)
- [OPH Hospital Policy Medicines Management](#)

- [OD 0647/16 National Standard for User-Applied Labelling of Injectable Medicines, Fluids and Lines 2016](#)
- NSQHS Standard 1,3,4,5,6,7,8

Standards, procedures, guidelines:

- [SCGH Nursing Practice Guideline No. 51 Medication Management](#)
- [SCGH Nursing Practice Guideline No. 4 Intravenous Therapy](#)
- [SCGH Patient Brochure Intravenous Iron Infusion](#)
- [OPH Nursing and Midwifery Practice Guidelines Intravenous Therapy](#)
- [OPH Surgical Services Practice Guidelines Day Procedure Unit Intravenous Iron Infusion](#)
- Intravenous (IV) Iron Infusions: Patient Information - [IronIV.pdf](#)

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